MATH PAPER 2

Compulsory Part · Multiple Choice

Past Paper Simulation (2025) — Paper 2

Paper 2 (MC)

75 minutes (1.25 hours)

FOR REVISION

For revision and self-assessment only. This is not an official examination paper.

Difficulty rating：**4.7** / 5.0

INSTRUCTIONS

1. This section contains multiple-choice questions; each has ONE correct answer.
2. All questions carry equal marks.
3. Choose only ONE answer for each question. More than one answer scores NO MARKS.
4. No marks will be deducted for wrong answers.

✶Independent, unofficial mock paper aligned to the DSE syllabus. Not affiliated with any official examination authority. For revision only.

There are 30 questions in Section A and 15 questions in Section B.
The diagrams in this paper are not necessarily drawn to scale. Choose the best answer for each question.

Section A

1. $\dfrac{(3^n)^2 \cdot 3^{-4}}{27^{n-1}} =$

A. $\dfrac{1}{3^{n+7}}$

B. $\dfrac{1}{3^{n+1}}$

C. 3^{5n-1}

D. 3^{n-1}

2. $16 - (x-3)^2 =$

A. $(1-x)(7+x)$

B. $(7-x)(x-1)$

C. $(7-x)(1+x)$

D. $(x+7)(x+1)$

3. If $(2x-a)(x+3) - b \equiv 2x^2 + x - 17$, where a and b are constants, then $b =$

A. -2

B. 17

C. 2

D. 32

4. If $a(x-2) = b(x+3)$, then $x =$

A. $\dfrac{2a-3b}{a-b}$

B. $\dfrac{2a+3b}{a-b}$

C. $\dfrac{a-b}{2a+3b}$

D. $\dfrac{2a+3b}{a+b}$

5. Let k be a constant. The roots of the equation $x^2 - (k+3)x + 3k = 0$ are

A. k or 3

B. $-k$ or -3

C. k or -3

D. $-k$ or 3

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6. If x is rounded off to 8.40 correct to 2 decimal places, then

A. $8.395 \leq x < 8.405$

B. \$8.395

C. $8.35 \leq x < 8.45$

D. $8.39 \leq x < 8.41$

7. The solution of the inequality $-7 \leq 2x + 1 < 9$ is

A. \$-4

B. $x < 4$

C. $-3 \leq x < 5$

D. $-4 \leq x < 4$

8. Let $f(x) = 2x^3 - kx^2 + 3x - 5$, where k is a constant. If $f(2) + f(-2) = -42$, then $k =$

A. -4

B. 8

C. 2

D. 4

9. Let $f(x) = x^3 + kx^2 - 4x + 12$, where k is a constant. If $x - 2$ is a factor of $f(x)$, then $f(-1) =$

A. 18

B. 12

C. 0

D. 6

10. A person deposits \$20 000 in a bank at an interest rate of 4% per annum, compounded half-yearly. Find the amount after 3 years, correct to the nearest dollar.

A. \$22 523

B. \$22 400

C. \$22 497

D. \$22 082

11. If $(a + b) : (b + c) : (c + a) = 5 : 6 : 7$, then $a : c =$

A. $3 : 4$

B. $4 : 3$

C. $5 : 7$

D. $2 : 3$

12. It is given that z varies directly as x^3 and inversely as y^2 . When $x = 2$ and $y = 3$, $z = 16$. When $x = 3$ and $y = 2$, $z =$

A. 81

B. 54

C. $\frac{243}{2}$

D. 32

13. In a sequence, $T(1) = 2$, $T(2) = 5$, and $T(n+2) = 2T(n+1) - T(n) + 3$ for any positive integer n . $T(5) =$

A. 32

B. 20

C. 47

D. 29

14. In the figure, the straight lines $L_1 : y = ax + b$ and $L_2 : y = cx + d$ intersect at a point with x-coordinate 2 . The y-intercept of L_1 is positive while that of L_2 is negative. Which of the following are true?
I. $aII. $b > d$
III. $a+b

A. I only

B. I and III only

C. II and III only

D. I and II only

15. The radius of the sector OAB is 6 cm and its perimeter is $(12 + 4\pi)$ cm. Which of the following are true?
I. The area of the sector is 12π cm^2 .
II. The length of the chord AB is $6\sqrt{2}$ cm.
III. $\angle AOB = 120°$

A. I only

B. I and II only

C. II and III only

D. I and III only

16. The height of a right circular cylinder is 8 cm and its total surface area is 66π cm^2 . Find the volume of a sphere whose radius is equal to the base radius of the cylinder.

A. 288π cm^3

B. 108π cm^3

C. 36π cm^3

D. 12π cm^3

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17. In the figure, $ABCD$ is a parallelogram and E is the mid-point of AB. DE produced and CB produced meet at F. If the area of $\triangle BEF$ is 3 cm^2, then the area of the quadrilateral $EBCD$ is

A. 12 cm^2
B. 6 cm^2
C. 9 cm^2
D. 15 cm^2

18. In the figure, D is a point on AC such that $AB^2 = AD \times AC$. If $AD = 4$ cm, $AC = 9$ cm and $BC = 7.5$ cm, then $BD =$

A. $\frac{10}{3}$ cm
B. 4.5 cm
C. 6 cm
D. 5 cm

19. In the figure, $ABCD$ is a square and $BCFE$ is a rhombus, where $\angle CBE = 120°$. Which of the following are true?
I. $AE = BF$
II. $\angle BAE = 15°$
III. $\triangle ABE \sim \triangle DCF$

A. I and II only
B. I and III only
C. I, II and III
D. II and III only

20. $ABCD$ is a trapezium, where $AD \parallel BC$, $AB = 10$ cm, $AD = 6$ cm, $\angle DAB = 120°$ and $\angle ADC = 135°$. Find BC correct to 3 significant figures.

A. 19.7 cm
B. 16.7 cm
C. 14.7 cm
D. 11.0 cm

21. In the figure, $ABCD$ is a rectangle, where $AB = 12$ cm and $BC = 5$ cm. Find the distance between D and AC.

A. $\frac{30}{13}$ cm
B. 5 cm
C. $\frac{60}{13}$ cm
D. $\frac{65}{12}$ cm

22. In the figure, $ABCDE$ is a pentagon inscribed in a circle, where $AE \parallel BD$ and BD bisects $\angle ABC$. If $\angle ADB = 30°$ and $\angle ABE = 20°$, then $\angle BCD =$

A. $100°$
B. $80°$
C. $50°$
D. $130°$

23. In the figure, $ABCD$ is a quadrilateral, where $\angle ABC = \angle BCD = 90°$, $AB = a$, $BC = b$ and $CD = c$, where $c > a$. $\tan \angle ADC =$

A. $\dfrac{c-a}{b}$
B. $\dfrac{b}{c-a}$
C. $\dfrac{b}{c+a}$
D. $\dfrac{a}{b}$

24. O is the pole. The polar coordinates of A and B are $(4, 30°)$ and $(4, 90°)$ respectively. The polar coordinates of C are $(r, 60°)$, where $r > 4$, such that $\triangle ABC$ is an equilateral triangle. $r =$

A. 8
B. $4\sqrt{2}$
C. $2\sqrt{3}$
D. $4\sqrt{3}$

25. Let P be a moving point on a plane such that its distance from a fixed point Q is always 5. The locus of P is

A. a straight line
B. a pair of parallel lines
C. a circle
D. a parabola

26. The straight lines $L_1 : 2x - y + 4 = 0$ and $L_2 : x + 2y + k = 0$ are perpendicular, where $k > 0$. If the area of the triangle bounded by L_1, L_2 and the x-axis is 20, then $k =$

A. 8
B. 22
C. 12
D. 2

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27. The centre of a circle is $(3, -4)$, and the chord cut by the x-axis on the circle has length 6. The equation of the circle is

A. $x^2 + y^2 + 6x - 8y = 0$
B. $x^2 + y^2 - 6x + 8y + 16 = 0$
C. $x^2 + y^2 - 6x + 8y - 11 = 0$
D. $x^2 + y^2 - 6x + 8y = 0$

28. A box contains 10 cards, 1 of which can be exchanged for $\$50$, 3 for $\$20$ each, and the rest for nothing. A card is drawn randomly from the box. Find the expected amount obtained.

A. $\$11$
B. $\$\frac{70}{3}$
C. $\$14$
D. $\$7$

29. The bar chart below shows the distribution of the numbers of days of exercise per week of 24 students in a class. Find the inter-quartile range of the distribution.

A. 3
B. 2
C. 1
D. 4

30. The mean and the range of the data 2, 5, 8, a, b are 6 and 10 respectively, where a and b are positive integers with $a<b$. The median is 5.
I. $b - a = 9$
II. The standard deviation is greater than 3.

A. I and II only
B. I, II and III
C. I and III only
D. II and III only

Section B

31. $2C5_{16} =$

A. $2^9 + 2^7 + 2^6 + 2^2 + 1$
B. $2^{10} + 2^8 + 2^7 + 2^3 + 2$
C. $2^9 + 2^8 + 2^6 + 2^2 + 1$
D. $2^9 + 2^7 + 2^5 + 2^2 + 1$

32. The L.C.M. of $x^2 - 4$, $x^3 - 8$ and $x^2 - 4x + 4$ is

A. $x - 2$

B. $(x-2)^3(x+2)(x^2+2x+4)$

C. $(x-2)^2(x+2)(x^2-2x+4)$

D. $(x-2)^2(x+2)(x^2+2x+4)$

33. The graph of $\log_3 y$ against $\log_9 x$ is a straight line with intercept 2 on the horizontal axis and intercept 4 on the vertical axis. If $y = kx^n$, where k and n are constants, then $n =$

A. -2

B. $-\frac{1}{2}$

C. -1

D. -4

34. Let a be a positive constant with $a \neq 1$. It is given that the graph of $y = \log_a x$ passes through the point $(4, -2)$. Which of the following are true?
I. $a > 1$
II. The graph of $y = a^x$ passes through the point $(2, 4)$.
III. The graph of $y = a^x$ and the graph of $y = \log_a x$ intersect.

A. III only

B. I only

C. II only

D. I and III only

35. $i + i^2 + i^3 + \cdots + i^{2027} =$

A. i

B. -1

C. $-1 + i$

D. 0

36. Let (x, y) be a point satisfying $x \geq 0$, $y \geq 0$, $x + 2y \leq 10$ and $3x + y \leq 15$. The greatest value of $2x + 3y$ is

A. 15

B. 10

C. 17

D. 18

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37. Let $T(n)$ be the nth term of an arithmetic sequence, where the common difference is non-zero and all terms are non-zero. Which of the following must be true?
I. $T(1)^2, T(2)^2, T(3)^2, \ldots$ is an arithmetic sequence.
II. $T(2), T(4), T(6), \ldots$ is an arithmetic sequence.
III. $\frac{1}{T(1)}, \frac{1}{T(2)}, \frac{1}{T(3)}, \ldots$ is a geometric sequence.

A. I only
B. II only
C. III only
D. II and III only

38. In the figure, AB is a diameter of the circle and TB is the tangent to the circle at B. AC produced meets TB at T. D is a point on the circle, and BD cuts AC at E. If $\angle CBT = 40°$ and $\angle ABD = 25°$, then $\angle AEB =$

A. $65°$
B. $105°$
C. $115°$
D. $125°$

39. How many roots does the equation $\tan^2\theta = 3\tan\theta$ have for $0° \le \theta \le 360°$?

A. 5
B. 4
C. 2
D. 3

40. $ABCD$ is a regular tetrahedron. Find the angle between the edge AB and the plane BCD, correct to the nearest degree.

A. $60°$
B. $71°$
C. $55°$
D. $35°$

41. The vertices of $\triangle OAB$ are $O(0,0)$, $A(6,0)$ and $B(0,b)$, where $b > 0$. If the x-coordinate of the in-centre of $\triangle OAB$ is 2, then the area of $\triangle OAB$ is

A. 48
B. 12
C. 30
D. 24

42. A group of 4 people is formed by selecting from 6 teachers and 5 students. If there is at least 1 teacher in the group, how many different groups can be formed?

A. 330
B. 720
C. 315
D. 325

43. A bag contains 5 red balls and 4 white balls. If 3 balls are drawn randomly from the bag at the same time, find the probability that at most 2 red balls are drawn.

A. $\frac{37}{42}$
B. $\frac{5}{42}$
C. $\frac{17}{42}$
D. $\frac{10}{21}$

44. The mean score of a test is 60 marks. Ka-yan scored 68 marks in the test and her standard score is 1. The score of Chi-ming differs from that of Ka-yan by 12 marks. The standard score of Chi-ming is

A. 2.5 only
B. 2.5 or -0.5
C. 1.5 or -1.5
D. -0.5 only

45. The mean, the range and the variance of the data $x_1, x_2, \ldots, x_n$ are m, R and V respectively. Consider the data $3x_1 - 2, 3x_2 - 2, \ldots, 3x_n - 2$. Which of the following are true?
I. The mean of the new data is $3m - 2$.
II. The range of the new data is $3R - 2$.
III. The variance of the new data is $3V$.

A. I and II only
B. I only
C. I and III only
D. II and III only

END OF PAPER